Compulsory Assignment 1

Predictive Analytics (KAN-CDSCO1005U)

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Question 1: GDP Level – Plot and ACF

```r
# install.packages("AER")
# install.packages("fpp3")
# install.packages("MetBrewer")

library(AER)
library(fpp3)
library(ggplot2)
library(MetBrewer)
```

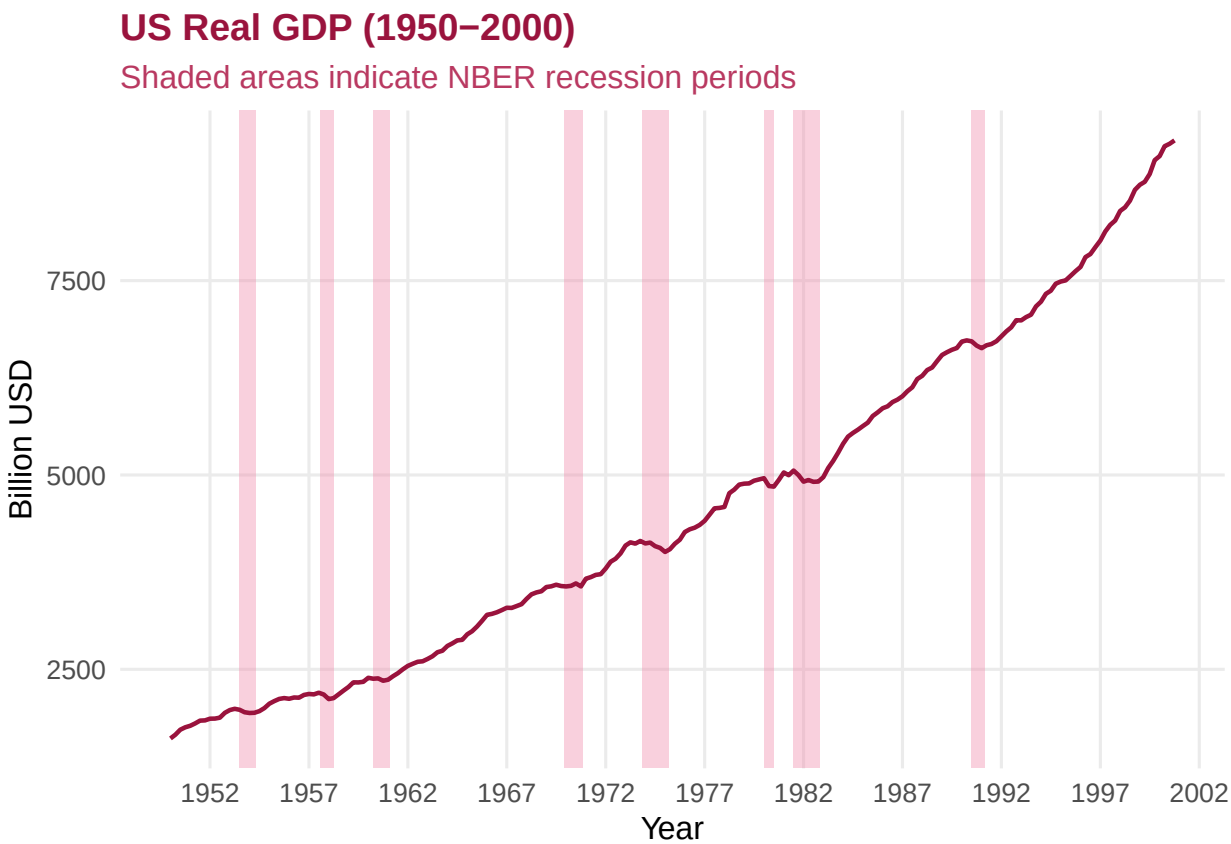
```r
data("USMacroG", package = "AER")
gdp <- USMacroG[, "gdp"]

# Load full Benedictus palette upfront for use throughout the assignment
bene <- met.brewer("Benedictus")

# Convert GDP to data frame for ggplot
gdp_df <- data.frame(
  date = as.Date(time(gdp)),
  gdp  = as.numeric(gdp)
)

# NBER recession periods within sample (1950-2000)
recessions <- data.frame(
  start = as.Date(c("1953-07-01", "1957-08-01", "1960-04-01", "1969-12-01",
                    "1973-11-01", "1980-01-01", "1981-07-01", "1990-07-01")),
  end   = as.Date(c("1954-05-01", "1958-04-01", "1961-02-01", "1970-11-01",
                    "1975-03-01", "1980-07-01", "1982-11-01", "1991-03-01"))
)
```

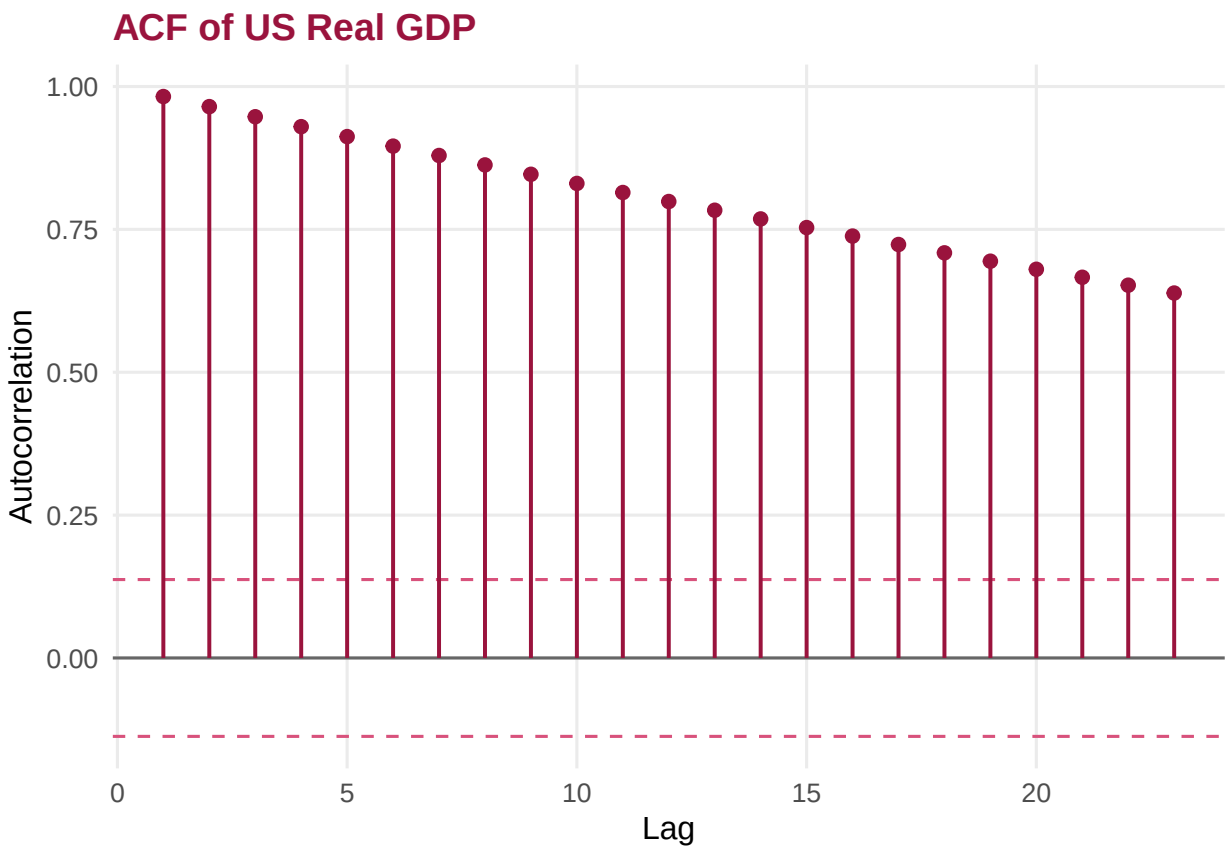
```r
ggplot(gdp_df, aes(x = date, y = gdp)) +
  geom_rect(data = recessions,
            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf),
            inherit.aes = FALSE,
            fill = bene[4], alpha = 0.4) +
  geom_line(color = bene[1], linewidth = 0.8) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  labs(
    title    = "US Real GDP (1950-2000)",
    subtitle = "Shaded areas indicate NBER recession periods",
    x        = "Year",
    y        = "Billion USD"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank()
  )
```



The GDP series displays a clear upward trend throughout the entire sample period from 1950 to 2000, indicating that the series is non-stationary in its level. There is no visible sign of seasonality, though brief slowdowns in growth are visible around the shaded recession periods, most notably in the early 1970s and early 1980s. The variance also appears to increase over time alongside the level of the series, which suggests that a log transformation may be appropriate before modelling.

```r
acf_data <- acf(as.numeric(gdp), plot = FALSE)
acf_df <- data.frame(
  lag = as.numeric(acf_data$lag[-1]),
  acf = as.numeric(acf_data$acf[-1])
)
ci <- qnorm(0.975) / sqrt(length(gdp))

ggplot(acf_df, aes(x = lag, y = acf)) +
  geom_hline(yintercept = 0, color = "grey40") +
  geom_hline(yintercept = c(-ci, ci), linetype = "dashed", color = bene[3]) +
  geom_segment(aes(xend = lag, yend = 0), color = bene[1], linewidth = 0.7) +
  geom_point(color = bene[1], size = 2) +
  labs(
    title = "ACF of US Real GDP",
    x     = "Lag",
    y     = "Autocorrelation"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    panel.grid.minor = element_blank()
  )
```



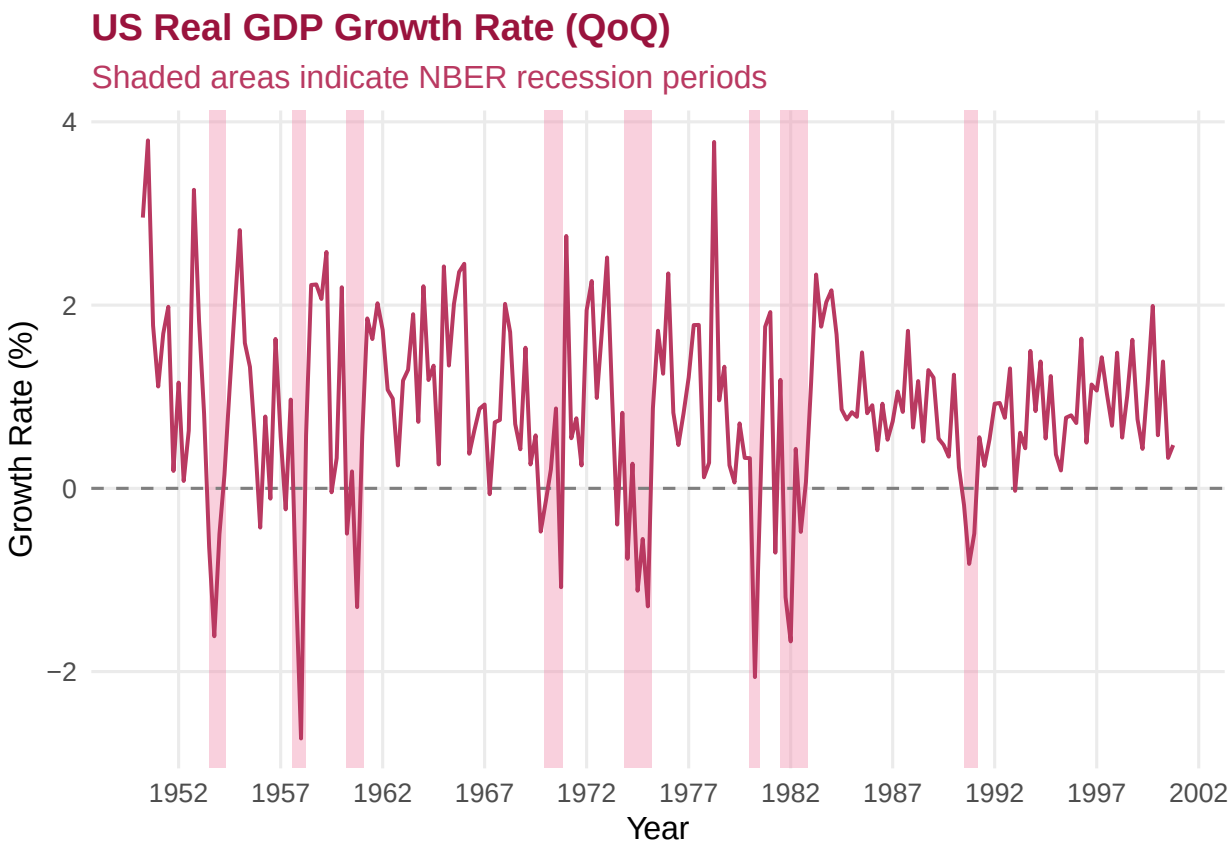
The ACF confirms the non-stationarity observed in the time series plot. Autocorrelations decay extremely slowly and remain highly significant across all displayed lags, which is a classic symptom of a unit root or a strong deterministic trend. This persistent autocorrelation structure further motivates transforming the data prior to any formal modelling, as most time series models require stationarity. A log transformation is warranted for two reasons: first, it stabilises the variance, which tends to grow proportionally with the level of GDP; second, differencing the log series yields approximate quarter-on-quarter percentage growth rates, which are economically meaningful and interpretable.

Question 2: GDP Growth Rate – Transformation, Plot, ACF, and Ljung-Box Test

```r
gdp_growth <- diff(log(gdp)) * 100  # Quarter-on-quarter log growth rate in %

gdp_growth_df <- data.frame(
  date   = as.Date(time(gdp_growth)),
  growth = as.numeric(gdp_growth)
)
```

```r
ggplot(gdp_growth_df, aes(x = date, y = growth)) +
  geom_rect(data = recessions,
            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf),
            inherit.aes = FALSE,
            fill = bene[4], alpha = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_line(color = bene[2], linewidth = 0.7) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  labs(
    title    = "US Real GDP Growth Rate (QoQ)",
    subtitle = "Shaded areas indicate NBER recession periods",
    x        = "Year",
    y        = "Growth Rate (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank()
  )
```

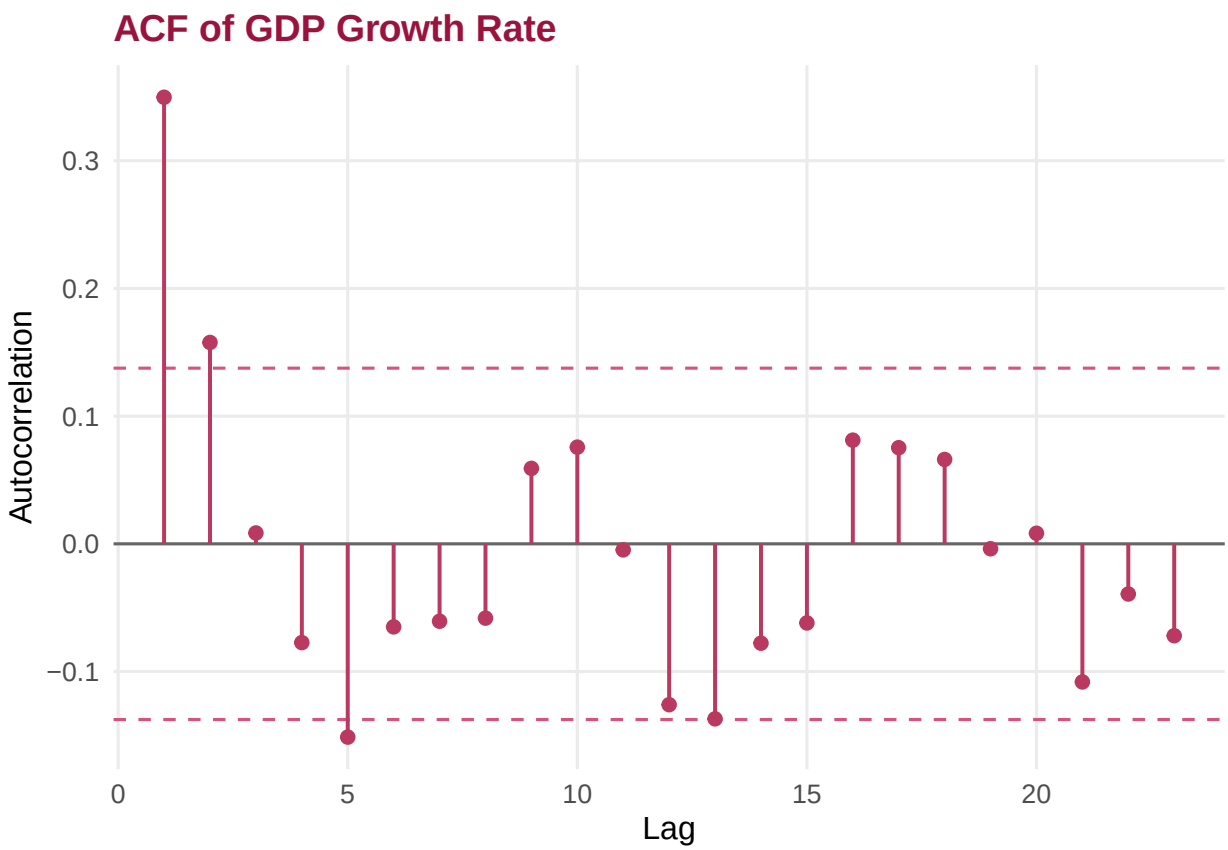


After taking the log-difference, the series no longer exhibits a trend – it fluctuates around a roughly constant mean close to zero, suggesting that differencing has successfully removed the non-stationarity present in the level series. There is no clear seasonal pattern, though some cyclical variation is visible, with more volatile and negative growth periods coinciding with the shaded recession episodes, most notably the sharp contractions of the early 1970s oil shock and the double-dip recession of 1980-1982. The variance appears broadly stable across the sample, supporting the appropriateness of the log transformation applied.

```
acf_growth <- acf(as.numeric(gdp_growth), plot = FALSE)
acf_growth_df <- data.frame(
  lag = as.numeric(acf_growth$lag[-1]),
  acf = as.numeric(acf_growth$acf[-1])
)
ci_growth <- qnorm(0.975) / sqrt(length(gdp_growth))

ggplot(acf_growth_df, aes(x = lag, y = acf)) +
  geom_hline(yintercept = 0, color = "grey40") +
  geom_hline(yintercept = c(-ci_growth, ci_growth),
             linetype = "dashed", color = bene[3]) +
  geom_segment(aes(xend = lag, yend = 0), color = bene[2], linewidth = 0.7) +
  geom_point(color = bene[2], size = 2) +
  labs(
    title = "ACF of GDP Growth Rate",
    x     = "Lag",
    y     = "Autocorrelation"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
```

```
    panel.grid.minor = element_blank()
  )
```



The ACF of the growth rate decays much more rapidly than that of the level series, which is consistent with a stationary process. A small number of lags appear statistically significant – notably lag 1 and potentially lag 3 – suggesting some remaining autocorrelation structure in the growth rate that may be relevant for model selection. The bulk of lags, however, fall within the confidence bands, indicating that the differenced log series behaves much closer to white noise than the original GDP series.

```
lb_test <- Box.test(gdp_growth, lag = 12, type = "Ljung-Box")

lb_tbl <- data.frame(
  Test = "Ljung-Box",
  Lags = 12,
  Statistic = round(unname(lb_test$statistic), 3),
  df        = unname(lb_test$parameter),
  p.value   = format.pval(lb_test$p.value, digits = 3, eps = 1e-3),
  check.names = FALSE
)

knitr::kable(
  lb_tbl,
  col.names = c("Test", "Lags", "X-squared", "df", "p-value"),
  caption   = "Ljung-Box test for GDP growth rate (H0: no autocorrelation up to lag 12).",
  align     = "lcccc"
)
```

Table 1: Ljung-Box test for GDP growth rate (H0: no autocorrelation up to lag 12).

Test	Lags	X-squared	df	p-value
Ljung-Box	12	44.262	12	<0.001

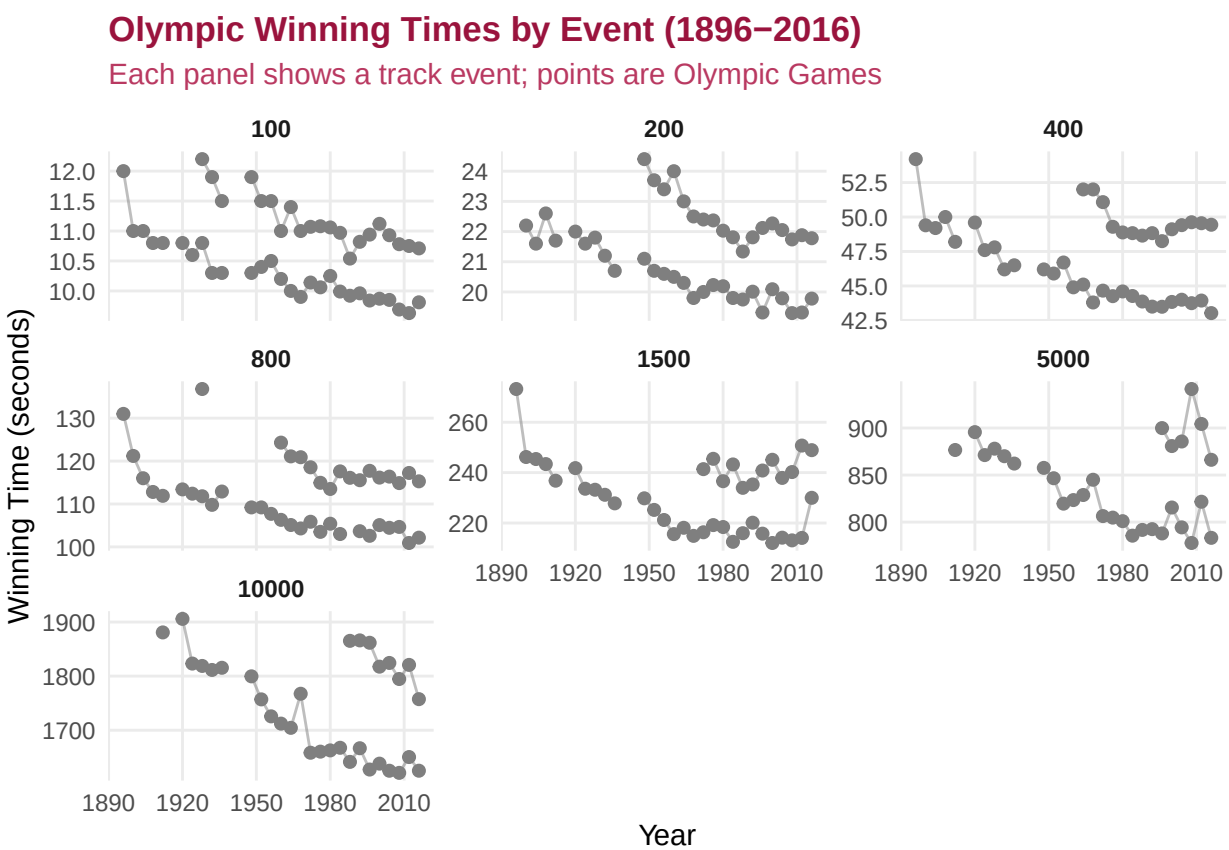
The Ljung-Box test evaluates the null hypothesis that the first 12 autocorrelations are jointly equal to zero, i.e. that the series is white noise. The test statistic is 44.262 with a p-value of 0. Since the p-value is below the conventional 5% significance level, we reject the null hypothesis and conclude that statistically significant autocorrelation is present in the GDP growth rate series. This finding is consistent with the ACF plot and implies that the growth rate is not pure white noise, which has important implications for forecasting – a model that exploits this autocorrelation structure should outperform a naive no-change forecast.

Question 3: Olympic Running

```r
data <- olympic_running
head(data)
```

```
## # A tsibble: 6 x 4 [4Y]
## # Key:       Length, Sex [1]
##    Year Length Sex    Time
##   <int>  <int> <chr> <dbl>
## 1  1896    100 men    12
## 2  1900    100 men    11
## 3  1904    100 men    11
## 4  1908    100 men    10.8
## 5  1912    100 men    10.8
## 6  1916    100 men    NA
```

```r
ggplot(data, aes(x = Year, y = Time, color = Sex)) +
  geom_point(size = 1.8) +
  geom_line(alpha = 0.5) +
  facet_wrap(~ Length, scales = "free_y", ncol = 3) +
  scale_color_manual(values = c("Men" = bene[1], "Women" = bene[5])) +
  labs(
    title    = "Olympic Winning Times by Event (1896-2016)",
    subtitle = "Each panel shows a track event; points are Olympic Games",
    x        = "Year",
    y        = "Winning Time (seconds)",
    color    = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank(),
    legend.position  = "bottom",
    strip.text       = element_text(face = "bold")
  )
```



The plots reveal several consistent features across all events. First, there is a clear downward trend in winning times over the course of the twentieth century, reflecting improvements in athletic training, nutrition, equipment, and race strategy. Second, men consistently outperform women across all distances, though women's events were introduced later and show steeper initial declines as the sport professionalised. Third, the rate of improvement appears to slow in more recent decades, suggesting diminishing returns as athletes approach physiological limits. Some events also show notable outliers, potentially reflecting unusual race conditions, doping disqualifications, or wars causing gaps in the data (1916, 1940, 1944).

```r
library(dplyr)

fits <- data %>%
  group_by(Length, Sex) %>%
  filter(!is.na(Time)) %>%
  summarise(
    slope = coef(lm(Time ~ Year))[2],
    .groups = "drop"
  )

fits_tbl <- fits %>%
  mutate(`Avg. change per year (sec)` = sprintf("%.4f", slope)) %>%
  select(Event = Length, Sex, `Avg. change per year (sec)`)

knitr::kable(
  fits_tbl,
  caption = "Estimated average change in winning time per year for each Olympic event.",
  align   = "llc"
)
```

Table 2: Estimated average change in winning time per year for each Olympic event.

Event	Sex	Avg. change per year (sec)	Year
100	men	NA	1896
100	men	NA	1900
100	men	NA	1904
100	men	NA	1908
100	men	NA	1912
100	men	NA	1920
100	men	NA	1924
100	men	NA	1928
100	men	NA	1932
100	men	NA	1936
100	men	NA	1948
100	men	NA	1952
100	men	NA	1956
100	men	NA	1960
100	men	NA	1964
100	men	NA	1968
100	men	NA	1972
100	men	NA	1976
100	men	NA	1980
100	men	NA	1984
100	men	NA	1988
100	men	NA	1992
100	men	NA	1996
100	men	NA	2000
100	men	NA	2004
100	men	NA	2008
100	men	NA	2012
100	men	NA	2016
100	women	NA	1928
100	women	NA	1932
100	women	NA	1936
100	women	NA	1948
100	women	NA	1952
100	women	NA	1956
100	women	NA	1960
100	women	NA	1964
100	women	NA	1968
100	women	NA	1972
100	women	NA	1976
100	women	NA	1980
100	women	NA	1984
100	women	NA	1988
100	women	NA	1992
100	women	NA	1996
100	women	NA	2000
100	women	NA	2004
100	women	NA	2008
100	women	NA	2012
100	women	NA	2016

Event	Sex	Avg. change per year (sec)	Year
200	men	NA	1900
200	men	NA	1904
200	men	NA	1908
200	men	NA	1912
200	men	NA	1920
200	men	NA	1924
200	men	NA	1928
200	men	NA	1932
200	men	NA	1936
200	men	NA	1948
200	men	NA	1952
200	men	NA	1956
200	men	NA	1960
200	men	NA	1964
200	men	NA	1968
200	men	NA	1972
200	men	NA	1976
200	men	NA	1980
200	men	NA	1984
200	men	NA	1988
200	men	NA	1992
200	men	NA	1996
200	men	NA	2000
200	men	NA	2004
200	men	NA	2008
200	men	NA	2012
200	men	NA	2016
200	women	NA	1948
200	women	NA	1952
200	women	NA	1956
200	women	NA	1960
200	women	NA	1964
200	women	NA	1968
200	women	NA	1972
200	women	NA	1976
200	women	NA	1980
200	women	NA	1984
200	women	NA	1988
200	women	NA	1992
200	women	NA	1996
200	women	NA	2000
200	women	NA	2004
200	women	NA	2008
200	women	NA	2012
200	women	NA	2016
400	men	NA	1896
400	men	NA	1900
400	men	NA	1904
400	men	NA	1908
400	men	NA	1912
400	men	NA	1920
400	men	NA	1924

Event	Sex	Avg. change per year (sec)	Year
400	men	NA	1928
400	men	NA	1932
400	men	NA	1936
400	men	NA	1948
400	men	NA	1952
400	men	NA	1956
400	men	NA	1960
400	men	NA	1964
400	men	NA	1968
400	men	NA	1972
400	men	NA	1976
400	men	NA	1980
400	men	NA	1984
400	men	NA	1988
400	men	NA	1992
400	men	NA	1996
400	men	NA	2000
400	men	NA	2004
400	men	NA	2008
400	men	NA	2012
400	men	NA	2016
400	women	NA	1964
400	women	NA	1968
400	women	NA	1972
400	women	NA	1976
400	women	NA	1980
400	women	NA	1984
400	women	NA	1988
400	women	NA	1992
400	women	NA	1996
400	women	NA	2000
400	women	NA	2004
400	women	NA	2008
400	women	NA	2012
400	women	NA	2016
800	men	NA	1896
800	men	NA	1900
800	men	NA	1904
800	men	NA	1908
800	men	NA	1912
800	men	NA	1920
800	men	NA	1924
800	men	NA	1928
800	men	NA	1932
800	men	NA	1936
800	men	NA	1948
800	men	NA	1952
800	men	NA	1956
800	men	NA	1960
800	men	NA	1964
800	men	NA	1968
800	men	NA	1972

Event	Sex	Avg. change per year (sec)	Year
800	men	NA	1976
800	men	NA	1980
800	men	NA	1984
800	men	NA	1992
800	men	NA	1996
800	men	NA	2000
800	men	NA	2004
800	men	NA	2008
800	men	NA	2012
800	men	NA	2016
800	women	NA	1928
800	women	NA	1960
800	women	NA	1964
800	women	NA	1968
800	women	NA	1972
800	women	NA	1976
800	women	NA	1980
800	women	NA	1984
800	women	NA	1988
800	women	NA	1992
800	women	NA	1996
800	women	NA	2000
800	women	NA	2004
800	women	NA	2008
800	women	NA	2012
800	women	NA	2016
1500	men	NA	1896
1500	men	NA	1900
1500	men	NA	1904
1500	men	NA	1908
1500	men	NA	1912
1500	men	NA	1920
1500	men	NA	1924
1500	men	NA	1928
1500	men	NA	1932
1500	men	NA	1936
1500	men	NA	1948
1500	men	NA	1952
1500	men	NA	1956
1500	men	NA	1960
1500	men	NA	1964
1500	men	NA	1968
1500	men	NA	1972
1500	men	NA	1976
1500	men	NA	1980
1500	men	NA	1984
1500	men	NA	1988
1500	men	NA	1992
1500	men	NA	1996
1500	men	NA	2000
1500	men	NA	2004
1500	men	NA	2008

Event	Sex	Avg. change per year (sec)	Year
1500	men	NA	2012
1500	men	NA	2016
1500	women	NA	1972
1500	women	NA	1976
1500	women	NA	1980
1500	women	NA	1984
1500	women	NA	1988
1500	women	NA	1992
1500	women	NA	1996
1500	women	NA	2000
1500	women	NA	2004
1500	women	NA	2008
1500	women	NA	2012
1500	women	NA	2016
5000	men	NA	1912
5000	men	NA	1920
5000	men	NA	1924
5000	men	NA	1928
5000	men	NA	1932
5000	men	NA	1936
5000	men	NA	1948
5000	men	NA	1952
5000	men	NA	1956
5000	men	NA	1960
5000	men	NA	1964
5000	men	NA	1968
5000	men	NA	1972
5000	men	NA	1976
5000	men	NA	1980
5000	men	NA	1984
5000	men	NA	1988
5000	men	NA	1992
5000	men	NA	1996
5000	men	NA	2000
5000	men	NA	2004
5000	men	NA	2008
5000	men	NA	2012
5000	men	NA	2016
5000	women	NA	1996
5000	women	NA	2000
5000	women	NA	2004
5000	women	NA	2008
5000	women	NA	2012
5000	women	NA	2016
10000	men	NA	1912
10000	men	NA	1920
10000	men	NA	1924
10000	men	NA	1928
10000	men	NA	1932
10000	men	NA	1936
10000	men	NA	1948
10000	men	NA	1952

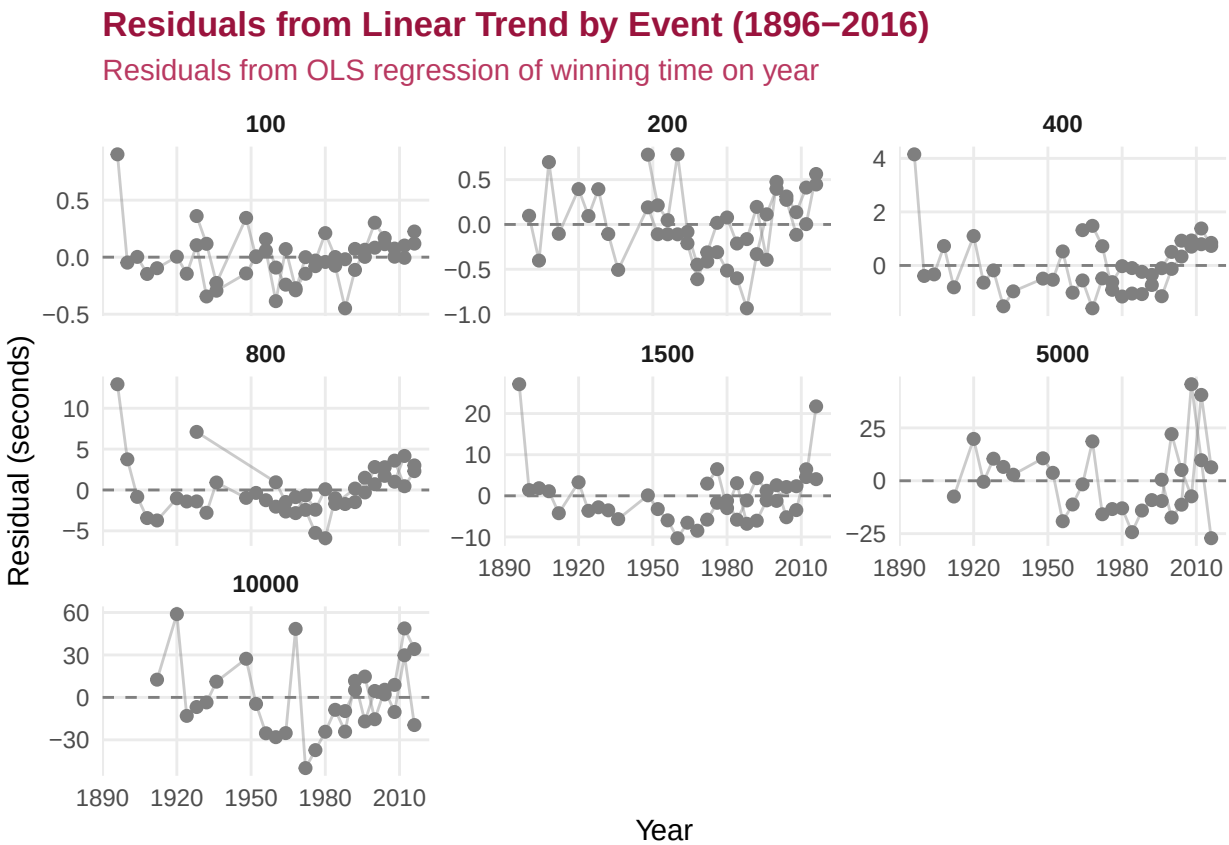
Event	Sex	Avg. change per year (sec)	Year
10000	men	NA	1956
10000	men	NA	1960
10000	men	NA	1964
10000	men	NA	1968
10000	men	NA	1972
10000	men	NA	1976
10000	men	NA	1980
10000	men	NA	1984
10000	men	NA	1988
10000	men	NA	1992
10000	men	NA	1996
10000	men	NA	2000
10000	men	NA	2004
10000	men	NA	2008
10000	men	NA	2012
10000	men	NA	2016
10000	women	NA	1988
10000	women	NA	1992
10000	women	NA	1996
10000	women	NA	2000
10000	women	NA	2004
10000	women	NA	2008
10000	women	NA	2012
10000	women	NA	2016

All slopes are negative, confirming that winning times have been decreasing across every event and for both sexes. The magnitude of the slope is largest for longer distance events in absolute terms, which is expected since longer races have more total time available for improvement. In relative terms, however, the rate of improvement is broadly similar across events. Women's events generally show steeper slopes, reflecting the fact that women's athletics was less developed at the start of the observation period and has seen larger absolute gains over time.

```r
data <- data %>%
  group_by(Length, Sex) %>%
  filter(!is.na(Time)) %>%
  mutate(
    fitted   = lm(Time ~ Year)$fitted.values,
    residual = lm(Time ~ Year)$residuals
  ) %>%
  ungroup()

ggplot(data, aes(x = Year, y = residual, color = Sex)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_point(size = 1.8) +
  geom_line(alpha = 0.4) +
  facet_wrap(~ Length, scales = "free_y", ncol = 3) +
  scale_color_manual(values = c("Men" = bene[1], "Women" = bene[5])) +
  labs(
    title    = "Residuals from Linear Trend by Event (1896-2016)",
    subtitle = "Residuals from OLS regression of winning time on year",
    x        = "Year",
    y        = "Residual (seconds)",
```

```
    color     = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank(),
    legend.position  = "bottom",
    strip.text       = element_text(face = "bold")
  )
```



The residual plots reveal systematic patterns that cast doubt on the suitability of a simple linear trend model. In most events, the residuals are not randomly scattered around zero – instead they tend to be positive in the early years, negative in the middle of the sample, and positive again towards the end. This curved pattern is consistent with a concave improvement trajectory: winning times improved rapidly in the early decades and have slowed in recent years as athletes approach physiological limits. A linear model cannot capture this curvature, suggesting that a nonlinear trend (e.g. a logarithmic or quadratic specification) may provide a better fit. Additionally, some events show clusters of large residuals around war years, reflecting the gaps in the data.

```
predictions <- data %>%
  filter(!is.na(Time)) %>%
  group_by(Length, Sex) %>%
  group_modify(~ {
    model <- lm(Time ~ Year, data = .x)
    pred  <- predict(model,
                     newdata  = data.frame(Year = 2020),
                     interval = "prediction",
```

```r
                      level    = 0.95)
    data.frame(
      Predicted = round(pred[1, "fit"], 2),
      Lower     = round(pred[1, "lwr"], 2),
      Upper     = round(pred[1, "upr"], 2)
    )
  }) %>%
  ungroup()

knitr::kable(
  predictions,
  col.names = c("Event", "Sex", "Predicted (sec)", "Lower 95% PI", "Upper 95% PI"),
  caption   = "Predicted winning times for the 2020 Olympics with 95% prediction intervals.",
  align     = "llccc"
)
```

Table 3: Predicted winning times for the 2020 Olympics with 95% prediction intervals.

Event	Sex	Predicted (sec)	Lower 95% PI	Upper 95% PI
100	men	9.53	9.03	10.04
100	women	10.53	10.02	11.04
200	men	19.12	18.39	19.85
200	women	21.20	20.02	22.38
400	men	42.04	39.55	44.53
400	women	48.44	45.87	51.01
800	men	99.24	91.78	106.70
800	women	111.48	103.41	119.56
1500	men	207.00	189.27	224.73
1500	women	245.46	232.19	258.72
5000	men	772.82	737.81	807.84
5000	women	892.11	782.40	1001.83
10000	men	1580.35	1515.75	1644.95
10000	women	1763.02	1706.69	1819.35

The predictions provide point estimates and 95% prediction intervals for the winning time in each event at the 2020 Tokyo Olympics. These forecasts rest on several important assumptions. First, the linear trend is assumed to continue into the future at the same constant rate observed historically – an assumption that is likely violated, as the residual analysis above suggests the rate of improvement has been slowing. This means the model will tend to produce optimistic (too fast) predictions, particularly for events where the curvature in residuals is most pronounced. Second, the model assumes that no structural breaks occur – such as new technological innovations, rule changes, or doping scandals – that would shift the trend. Third, the prediction intervals are derived under the assumption of normally distributed, homoskedastic residuals; if this does not hold, the intervals may be too narrow or too wide. In practice, the actual 2020 results can be compared against these predictions to evaluate the model's out-of-sample performance.